

# Community Resilience: Quantifying the Disruptive Effects of Sudden Spikes in Activity within Online Communities

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## ABSTRACT

Online communities often experience some form of crisis. Regardless of whether these communities are small groups or entire platforms, the ability to handle disruptions during volatile periods signifies the resilience of a community. In this paper, we analyze the effects of a specific type of crisis event on Reddit: sudden spikes in attention received by a community due to a post from the subreddit hitting r/popular—the default feed that aggregates the most popular content on Reddit. Our results show that r/popular is a source of potential disruptions due to the higher number of comments, authors, and especially removed comments found compared to threads that do not reach r/popular. When looking at r/popular’s effects across subreddits, large communities’ commenting and posting behaviors were less impacted by r/popular threads compared to smaller communities. However, similar-sized subreddits had substantial variations in how their commenting and posting activities were influenced by an r/popular thread. Understanding the causal factors behind these differences has implications for online governance, as well as fostering resilient and healthy communities.

## CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Social media**.

## KEYWORDS

online communities, Reddit, moderation, resilience

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## 1 INTRODUCTION

Online communities have become increasingly integral to our day-to-day lives, from the way we socialize and connect with one another to how we learn about ongoing events. Given the rising popularity of virtual spaces, it is especially important to understand what makes an online community successful as well as how to reach that level of success. When evaluating what makes a successful online community, the first things we often think about are metrics like growth and quality of content [24]. However, taking a deeper look into the literature we find other important, but more subjective features like the ability to integrate new users [16, 17] and maintain communal norms [7]. Even though we know that these traits are valuable, communities of all sizes still experience crises: events that severely disrupt the activities of an online community [16, 21]. These crises can be the result of organized trolling campaigns [10, 18], prominent moderator interventions [6, 21], and sudden spikes in activity [16]. No matter the origin, the pervasiveness of crises across the web illustrates how crucial it is for a community to be able to withstand the impact of crises.

In this paper, we examine multiple Reddit communities and their respective ability to withstand crises to develop a concept called *community resilience*. Based on prior work [3, 8, 20], we define resilience as *the ability to reduce the impact of disruptions during crises*. Understanding community resilience and the factors that influence it has practical implications for how platforms and moderators govern their communities and provide stability for their users. Instead of relying solely on reactive approaches to combat disruptive and antisocial behavior after they occur, communities can adopt preventative approaches to foster resilience in a proactive manner [14]. Moreover, the notion of community resilience serves as a natural extension to prior work on defining, identifying, and facilitating successful online communities [16, 17, 19, 24]. For example, the lack of antisocial behavior does not necessarily indicate a thriving and successful community. Instead, a successful community might be characterized by its ability to withstand disruptive behavior, and bounce back in the face of adversity.

To examine the resilience of communities, we look towards Reddit and “crises” generated by appearing or “hitting” its default (i.e., popular) feed called r/popular. Reddit is a forum-based social media platform that hosts communities, called subreddits, that are often regulated by volunteer moderators. Based on a ranking algorithm that combines high-volume of upvotes and recency, Reddit

compiles a dynamic collection of “popular” threads from across different subreddits and these popular threads collectively appear on r/popular. Threads on r/popular—and subsequently the subreddits that generated these threads—are exposed to millions of new users browsing the popular feed. These new sets of eyes, often unfamiliar to the rules that govern the space, have the potential to discuss and contribute within these subreddits leading to disruptions if not properly integrated [16, 17]. The popular feed will serve as the origin for crises that we analyze in this work because it can lead to substantial spikes in activity within subreddits. Examining how subreddits react to such spikes in activity will allow us to understand community-level differences in response and resilience when faced with disruptions. More specifically, we ask the following research questions about hitting r/popular and how it impacts subreddits:

- (RQ1) What impact does hitting r/popular have on Reddit posts?
- (RQ2) How are different communities affected when their posts hit r/popular?
- (RQ3) What are causal factors that affect the intensity and longevity of r/popular’s impacts on communities?

We answer these questions by collecting posts (referred to by threads here on) and comments from subreddits that hit r/popular between February 23, 2020 and July 30, 2020. Our final dataset comprises of a total of 7,139 r/popular threads from 30 study subreddits with varying number of subscribers, along side hundreds of thousands of control threads (i.e., threads that did not hit r/popular) and millions of comments generated within these communities. Using this dataset, we employ statistical methods, including Wilcoxon signed-rank tests using matched pairs (Section 4.1) and linear models (specifically Poisson regressions in Section 4.2) to estimate the thread-level and community-level effects of hitting r/popular within the study subreddits. Through our analysis, we find that at a thread-level, threads that hit r/popular receive higher levels of activity, newcomer participation and comment removals (Section 4.1). At the community-level, we find spill-over effects caused by an individual thread from a subreddit hitting r/popular, observing significant spikes in overall activity levels and newcomer influx within the subreddit. We find that not all communities are affected in the same way: the number of comments and threads from large subreddits are less impacted by individual r/popular threads when compared to small and medium subreddits. Likewise, similar-sized subreddits exhibit differential effects from r/popular threads (Section 4.2). Examining the causal factors leading to such variations in intensity and longevity of impacts from sudden spikes in activity will allow us to understand what makes communities resilient to crises.

Next, we review existing work, and then present our empirical investigation into answering RQ1 and RQ2. Finally, we reflect on some of the opportunities introduced by findings from our empirical work so far (e.g., RQ3), and conclude by discussing our plans to address these questions.

## 2 BACKGROUND

To situate this work, we will highlight some relevant literature in three areas: (1) understanding what online community success means, (2) measuring crises in online communities, and (3) defining community resilience.

**Community Success.** There has been extensive literature that discusses what makes successful online communities [9, 17, 24]. Kraut and Resnick [17] is a particularly seminal work in the area as it highlights numerous challenges that come with building successful online communities. This includes encouraging content contributions, membership commitment, and integrating newcomers. Despite their work, a significant portion of the literature in community success often focuses on growth metrics [11, 15, 16, 26]. Cunha et al. [9] serves as a recent deviation from this paradigm as it highlights additional avenues for communities to be successful; avenues that are distinct in that they are not highly correlated. Not only are there different success pathways, Weld et al. [24] found that community members and moderators differ in what they consider community improvements (e.g., community members valued democracy more than moderators). Additionally, there have been other articles that focus on attributes like membership overlap [26], content quality [19], and sense of community [16]. All of these examples illustrate the complexity of community success because it incorporates multiple different features and perspectives. Our work builds upon the community success literature by analyzing a community’s ability to withstand crises (i.e., community resilience).

**Community Crisis.** All of this work in understanding community success is in part trying to reduce instability and antisocial behavior that communities experience, whether those communities are small groups [16] or entire platforms like Reddit [6, 10, 18, 21–23] and Twitter [13, 25]. These papers often analyze antisocial behavior through detection [4] and prevention through moderator interventions and tools [5, 6, 13]. In contrast to antisocial behavior, community health is often viewed in the negative sense of avoiding bad behavior in online communities [2]. However, we believe that focusing on developing prosocial measures and strategies have substantial potential in curbing these negative behaviors. Bao et al. [1] also shares this interest in their work analyzing prosocial aspects of conversations online. With sufficient development, we believe that community resilience will also belong in the category of prosocial metrics.

**Community Resilience.** A significant inspiration for our work comes from Butler et al. [3] where they defined community resilience as the ability for community membership to be stable in spite of topical changes. Although important, we modify it to instead focus on crises which are often sudden, unexpected, and highly disruptive. This viewpoint matches the definition specified in Cohen et al. [8], but there are a variety of other ways to interpret community resilience as it encompasses factors like community adaptability, agency, vulnerability, and so on. Those variations and distinctions are described thoroughly in Matarrita-Cascante et al. [20]. For the rest of this paper, we will only focus on crisis management when discussing community resilience.

## 3 DATA

As mentioned earlier, we are interested in examining the impact of a community thread reaching r/popular<sup>1</sup>—the default feed that aggregates the most popular (i.e., the most highly voted) content across subreddits. Similar to other subreddits, r/popular is a dynamic feed that updates over time to display the most voted-on and active

<sup>1</sup><https://www.reddit.com/r/popular/>

content on the platform. The feed updates in real time and serves only as a content aggregator, thus no records exist for r/popular. As a result, the data needed to be collected in real time. To do this, we collected the top 100 r/popular threads every six hours from February 23, 2020 to July 30, 2020 using the Python Reddit API Wrapper (PRAW).<sup>23</sup> The resulting dataset contained 43,557 r/popular threads spanning across 883 subreddits. For tractability, the 883 subreddits observed were split into three equal-sized groups (large, medium, and small subreddits) based on the number of subscribers each community had. For each group, we collected comments and threads from ten randomly-selected subreddits totaling to 30 study subreddits. The subreddit comment and thread data were collected during the same date range as the r/popular dataset, in addition to a few months prior to the initial date (i.e., July 30, 2019 to July 30, 2020) using the Python Pushshift.io API Wrapper (PSAW).<sup>4</sup> A summary of the dataset we analyzed is provided in Table 1.

## 4 ANALYSIS & FINDINGS

Here we will present the statistical methods, analysis, and interpretation for each research question, starting with the most fundamental research question that establishes that the differences between r/popular threads and threads that do not reach r/popular.

### 4.1 Thread-Level Effects of Hitting r/popular (RQ1)

**Wilcoxon Signed-Rank Test.** To determine whether r/popular had effects on individual subreddits, we used a one-way Wilcoxon signed-rank test using matched pairs to test whether r/popular threads (i.e., treatment threads) had (1) a greater number of comments, (2) a higher percentage of removed comments, and (3) a greater number of distinct authors compared to threads that did not make it to r/popular (i.e., control threads). We examined these three metrics because spikes in these measures are often disruptive if not handled properly [16, 22, 23]. Additionally, we used a Wilcoxon signed-rank test because we could not assume that these three metrics fell on a normal distribution—hence a non-parametric test was used to be conservative. The small subreddits were omitted from this test due to insufficient r/popular threads, so further conclusions only apply to medium and large subreddits.

**Propensity Score Matching.** To generate treatment-control pairs for the Wilcoxon signed-rank test, we used propensity score matching [12] to find control threads that were most similar to the treatment thread so that the comparisons were more valid than picking a random thread. The propensity score matching algorithm used the following covariates to pair treatment and control threads: (1) the Unix timestamp of when the thread was created; (2) the domain of the link attached on the thread; (3) whether the thread contains media hosted on Reddit; (4) whether the thread contains a video; (4) whether the thread contains content restricted to adults (i.e., over 18); (5) the length of the text contained in the thread (0 if no thread text); and (6) the length of the title. These features were selected over others because they can be used to represent a thread's

| Subreddit         | #popular | #comments  | #threads |
|-------------------|----------|------------|----------|
| politics          | 896      | 12,520,845 | 197,384  |
| pics              | 818      | 4,610,057  | 281,361  |
| aww               | 748      | 3,008,561  | 462,008  |
| funny             | 707      | 4,238,608  | 425,158  |
| worldnews         | 666      | 6,059,124  | 225,099  |
| todayilearned     | 660      | 2,795,024  | 71,573   |
| nextfuckinglevel  | 615      | 2,062,885  | 45,186   |
| interestingasfuck | 610      | 2,421,460  | 63,198   |
| news              | 603      | 5,160,540  | 363,689  |
| gaming            | 596      | 3,418,176  | 303,337  |

(a) Large Subreddits

| Subreddit            | #popular | #comments | #threads |
|----------------------|----------|-----------|----------|
| suicidebywords       | 22       | 53,844    | 6,219    |
| FunnyandSad          | 19       | 60,395    | 8,518    |
| DesignPorn           | 19       | 52,977    | 3,461    |
| thatHappened         | 19       | 180,547   | 16,288   |
| witcher              | 19       | 230,560   | 18,839   |
| Competitiveoverwatch | 19       | 536,046   | 12,645   |
| lgbt                 | 19       | 475,919   | 57,293   |
| community            | 18       | 162,293   | 18,098   |
| cyberpunkgame        | 18       | 266,438   | 12,912   |
| PraiseTheCameraMan   | 18       | 50,908    | 2,771    |

(b) Medium Subreddits

| Subreddit             | #popular | #comments | #threads |
|-----------------------|----------|-----------|----------|
| tattoos               | 3        | 113,323   | 22,765   |
| PersonalFinanceCanada | 3        | 475,299   | 31,221   |
| oddlyspecific         | 3        | 48,552    | 7,207    |
| Instantregret         | 3        | 68,635    | 4,330    |
| thisismylifenow       | 3        | 18,113    | 1,717    |
| Baking                | 3        | 179,305   | 37,872   |
| startrek              | 3        | 261,830   | 10,436   |
| BattlefieldV          | 3        | 371,947   | 30,111   |
| comicbooks            | 3        | 188,877   | 17,843   |
| OldPhotosInRealLife   | 3        | 27,226    | 1,361    |

(c) Small Subreddits

**Table 1: Number of popular threads, comments, and threads collected for each subreddit between February 23, 2020 to July 30, 2020.**

characteristics at the time of posting and remain static (i.e., they do not change over time). From inspection, the algorithm heavily prioritized temporal proximity so the pairs were often posted within the same day or hour. An example of a matched pair can be found in Appendix A2.

**Findings from Wilcoxon Signed-Rank Test.** The Wilcoxon signed-rank tests showed that r/popular threads have a greater number of comments and unique authors compared to threads from the same subreddit that did not reach r/popular. This result holds for all large and medium subreddits analyzed. The differences, expressed by the resulting coefficients, were especially drastic for the large subreddits compared to medium subreddits. That makes intuitive

<sup>23</sup><https://praw.readthedocs.io/en/stable/>

<sup>4</sup>During collection, there was a three-day down period from April 14, 2020 to April 16, 2020.

<sup>4</sup><https://psaw.readthedocs.io/en/latest/>

sense as the large subreddits often have threads reach higher levels of peak activity compared to medium subreddits. Regarding the rate of comment removals, all of the large subreddits' r/popular threads had a higher percentage of removed comments compared to control threads, but there were non-statistically significant results for some of the medium subreddits. A potential explanation could be that having smaller moderation teams leads to a lack of comment removals within medium subreddits. Although these tests conclude that r/popular threads and control threads are different in these measures, it does not tell us whether hitting r/popular was the cause for these differences. Nevertheless, these differences show that r/popular threads can be a source of disruption due to the high levels of activity generated.

## 4.2 Variations in Subreddit-Level Effects of Reaching r/popular (RQ2)

**Poisson Regression Analysis.** To estimate subreddit-level effects of hitting r/popular, we computed the daily number of r/popular threads within a subreddit and examined its relation to six community metrics that were also computed daily: (1) the number of comments, (2) the number of unique comment authors, (3) the number of *new* comment authors who have not posted before on the subreddit—we looked back as far as July 30, 2019, (4) the number of removed comments, (5) the number of threads, (6) the number of unique thread authors each day. Essentially, these six daily community metrics form a time series (an example is shown in Appendix A1) that can be compared to the aggregate number of r/popular threads to estimate the relationship between daily aggregate number of r/popular threads and the metrics described above. We used these six easy-to-interpret community metrics to explore the potential impacts of hitting r/popular on communities. These metrics are not exhaustive, and future work should explore other, more complex community metrics—for example, daily toxicity levels provided by the Perspective API<sup>5</sup> and so on.

With the generated community metrics, we ran pairwise tests with generalized linear models, specifically Poisson regressions given that our outcome variables of interest were raw counts. The Poisson regression analyses give us the ability to compute the percentage change an r/popular thread has on a subreddit's daily number of comments as well as the other five metrics we calculated. Table 2 displays the results from our regression analyses.

**Findings from Regression Analyses.** Results from our regression analysis demonstrates the spill-over effects of individual r/popular threads on overall activity within subreddits. Almost all study subreddits had significant, positive associations between hitting r/popular and community metrics like activity levels, newcomer influx, and comment removals. For instance, we found that an increase in the number of r/popular threads on a given day within all study subreddits—large, medium and small—resulted in a significant increase in the number of newcomers posting within the subreddit. With the exception of two medium and two small subreddits, we also observed that an increase in r/popular threads corresponds to an increase in the number of comment removals within the subreddits. In other words, hitting r/popular generates significant spikes in activity levels and newcomer participation

within subreddits, thereby having the potential leading to more comment removals and other undesirable behavior.

*Differences in Effects Across Subreddit Groups.* We found that large subreddits are less impacted by a single r/popular thread compared to medium and small communities (as observed through percent changes in metrics shown in Table 2). One potential reason for this trend could be the sheer number of r/popular threads in these large subreddits making any one individual thread have minimal impacts on the overall community. Another explanation could be that such crises are commonplace within larger subreddits, thereby reducing the impact of individual r/popular threads. This indicates that increased exposure to crises in the past could play a role in a community's resilience to the impacts of a new crisis.

*Differences in Effects Within Subreddit Groups.* Despite not having massive percentage changes when compared to medium and small subreddits, large subreddits do exhibit variations in how they are affected by individual r/popular threads. When looking within the group of large subreddits, r/pics and r/news stand out from the rest, experiencing larger increases in activity levels upon hitting r/popular. The characteristics that led to these within-group differences have yet to be explored, but understanding these reasons could lead to recommendations for communities to handle the impacts of elevated levels of activity and new user influx. This is especially important because r/pics and r/news also exhibited a higher percentage of removed comments because of r/popular threads—indicating that there was an increase in the number of norm violations caused by threads hitting r/popular.

Similar within-group differences are observed in the case of medium and small subreddits as well—for example, r/lgbt (least increase in #comments) and r/cyberpunkgame (most increase in #comments) compared to the rest of the subreddits in the group (as shown in Table 2). Another potential reason for communities exhibit varying levels of effects from hitting r/popular could be the nature of content and discussions (e.g., how accessible is the subreddit's topic of discussion to newcomers). Let us consider the case of r/PersonalFinanceCanada which is a small subreddit, so we would expect r/popular to have a substantial impact on activity levels within the subreddit. However through our analysis, we found that is not really the case, specifically with regards to increase in number of comments (14.79%) and authors (11.41%). The reason for these small spikes (when compared to other small subreddits) could be the expertise required to contribute within this community or perhaps the moderation team mandating (and enforcing) certain guidelines on comments posted within the subreddit, thus making r/PersonalFinanceCanada more resilient to sudden spikes in attention.

*Contextual Differences.* However, some variations (within and across subreddit groups) may not necessarily be indicative of community resilience, but rather contextual differences. For example, the variation between r/community and r/cyberpunkgame, both similar-sized communities, is drastic—r/community sees a 56.02% increase in the number of comments compared to r/cyberpunkgame's 152% increase, when a post hits r/popular. The most likely reason for this difference is the following: the growth in activity on r/community (a subreddit dedicated to a television show) lasted throughout the show's airtime whereas r/cyberpunkgame (a subreddit dedicated to a video game) rose rapidly with the release of

<sup>5</sup><https://www.perspectiveapi.com/>

**Table 2: The pairwise Poisson regression results comparing daily r/popular threads for each subreddit to the community metrics described in Section 4.2 converted to percentage change. For example, each r/popular thread results in a 3.3% increase in comments in r/politics. This table highlights the variations in r/popular thread impacts across different-sized communities and the overall trend is that larger subreddits are less affected by r/popular threads. Empty cells were not statistically significant ( $\alpha = 0.05$ ). Bonferroni corrections were applied.**

| Subreddit             | Comments  |          |              | Threads   |          |          |
|-----------------------|-----------|----------|--------------|-----------|----------|----------|
|                       | #comments | #authors | #new authors | #removals | #threads | #authors |
| politics              | 3.3%      | 2.13%    | 3.12%        | 4.15%     | 2.82%    | 2.07%    |
| pics                  | 15.58%    | 11.42%   | 10.08%       | 15.32%    | 5.1%     | 2.89%    |
| aww                   | 0.63%     | 0.86%    | 2.26%        |           | 0.77%    | 0.56%    |
| funny                 | 2.64%     | 2.65%    | 4.6%         | 2.86%     | 2.5%     | 3.01%    |
| worldnews             | 3.48%     | 4.23%    | 8.16%        | 7.31%     |          | 1.85%    |
| todayilearned         | 2.2%      | 2.21%    | 2.23%        | 4.81%     | 2.55%    |          |
| nextfuckinglevel      | 10.23%    | 8.56%    | 8.09%        | 14.82%    | 7.56%    | 4.99%    |
| interestingasfuck     | 9.77%     | 8.52%    | 7.3%         | 11.57%    | 5.32%    | 2.47%    |
| news                  | 17.04%    | 13.41%   | 12.32%       | 17.4%     | 7.09%    | 5.0%     |
| gaming                | 2.5%      | 2.49%    | 3.08%        | 4.33%     |          |          |
| suicidebywords        | 84.13%    | 71.62%   | 103.37%      |           |          |          |
| FunnyandSad           | 132.82%   | 96.93%   | 149.36%      | 250.59%   |          |          |
| DesignPorn            | 70.0%     | 59.19%   | 88.49%       | 168.6%    |          |          |
| thatHappened          | 19.39%    | 20.34%   | 30.97%       |           |          |          |
| witcher               | 27.39%    | 30.41%   | 45.05%       | 51.42%    | 9.27%    | 10.66%   |
| Competitiveoverwatch  | 62.93%    | 38.7%    | 78.79%       | 47.68%    | 42.21%   | 40.24%   |
| lgbt                  | 14.44%    | 13.59%   | 12.1%        | 56.05%    | 10.04%   | 6.53%    |
| community             | 56.02%    | 51.74%   | 61.25%       | 57.76%    | 38.3%    | 33.0%    |
| cyberpunkgame         | 152.0%    | 126.04%  | 179.99%      | 168.09%   | 150.93%  | 129.8%   |
| PraiseTheCameraMan    | 49.14%    | 48.89%   | 65.63%       | 197.9%    |          |          |
| tattoos               |           |          | 21.84%       | 66.6%     |          |          |
| PersonalFinanceCanada | 14.79%    | 11.41%   | 70.82%       |           | 30.47%   | 37.9%    |
| oddlyspecific         | 303.37%   | 247.36%  | 430.12%      | 726.67%   |          |          |
| Instantregret         | 94.09%    | 43.58%   | 67.53%       |           |          |          |
| thisismylifenow       | 60.89%    | 58.16%   | 84.73%       | 155.71%   |          |          |
| Baking                | 52.82%    | 62.17%   | 155.2%       | 290.61%   | 24.48%   | 22.29%   |
| startrek              | 76.9%     | 78.41%   | 194.37%      | 59.26%    | 27.33%   |          |
| BattlefieldV          | 216.4%    | 197.93%  | 545.49%      | 189.98%   | 174.49%  | 170.95%  |
| comicbooks            | 37.38%    | 19.2%    | 89.23%       | 149.26%   |          |          |
| OldPhotosInRealLife   | 405.15%   | 377.05%  | 517.68%      | 665.89%   |          |          |

the game. This contrast can be seen in Appendix A3. An example where context might not matter as much would be r/worldnews and r/news which share similar types of content, but r/news had 17.04% increase in comments for each r/popular thread compared to r/worldnews with 3.48%. The differences between r/worldnews and r/news extends to the other metrics as well (excluding the non-significant association for number of threads on r/worldnews). Overall, we found that threads hitting r/popular had a less intense impact on r/worldnews when compared to r/news—indicating that r/worldnews might be more resilient than r/news based on the intensity of effects following a crisis.

### 4.3 Factors that Affect the Intensity and Longevity of Community-level Effects (RQ3)

As seen in Table 2, there is a significant amount of variation in the disruptive, spillover effects of r/popular threads on subreddits, both within and across groups. Additionally, our previous regressions were unable to capture the variations between r/popular threads within the same subreddit. In other words, what causes certain r/popular threads to result in more drastic effects compared to others? To investigate this question, we will examine differences between the community-level effects of r/popular threads through two lens: longevity and intensity. To do so, we will estimate the magnitude of the effects of r/popular threads on the community (i.e., intensity) and how long these effects last (i.e., longevity).

To illustrate, consider the example of newcomer influx (or #new authors in Table 2) to a community as a result of threads hitting

r/popular. We plan to estimate intensity by measuring the *magnitude* of change in the number of newcomers brought into a community resulting from a given r/popular thread—#new authors posting within the popular thread. Next, we plan to estimate how long the newcomers brought into the community through the r/popular thread continue to post on other parts of the subreddit. Specifically, at what time interval  $t$  after the r/popular thread “ends” (i.e., timestamp when the last comment was made within the popular thread), do the *majority* of newcomers stop participating within the community? This example demonstrates how we plan to examine the longevity and intensity of newcomer influx resulting from threads hitting r/popular. Using similar formulations, we will examine a range of community-level effects beyond newcomer influx, including activity levels, norm violations, and removal rates.

We plan to use statistical methods to explore correlations between longevity and intensity of effects caused by r/popular threads and features of r/popular threads (e.g., content type, topic, and controversiality), authors (e.g., karma, account age, and posting history of the authors of r/popular threads), and subreddits (e.g., subscriber counts, average activity and removal rates, size of moderation team) to answer RQ3. These subreddit-, thread-, and author-level associations will shed light onto what characteristics impact the community-level effects of threads hitting r/popular, and thereby identify factors that contribute to community resilience in the face of sudden spikes in activity.

## 5 DISCUSSION

To recall, our goal was to understand how communities handle crises—in the form of sudden spikes in external attention and activity—when threads hit r/popular. By examining variations in community-level impacts of such crises we presented a notion of community resilience, thus building upon prior definitions of successful online communities. In this paper, we performed exploratory analyses on the effects of hitting r/popular and established that r/popular threads received a higher number of comments, authors as well as a higher percentage of removed comments when compared to threads that do not reach r/popular.

From a community perspective, r/popular threads impacted not only the number of comments and threads, but other features like new users and comment removals. When looking across different-sized communities, we found that large subreddits are impacted less by individual r/popular threads compared to smaller communities. Despite this overall trend, there still exists some unexplained variations between similar-sized communities (e.g., r/news and r/worldnews) that have the potential to be attributed to the resilience of communities. The resilience exhibited by a community in the face of adversity (measured in terms of longevity and intensity of effects) could be the result of a range of factors like exposure to crises in the past, existence of community guidelines and moderators, topics of discussion, and so on.

In future work, we plan to examine potential causal factors that influence a community’s resilience and ability to rebound from crisis events. We can start by examining the features highlighted by prior work discussing how an online community managed a surge of newcomers [16], namely: a strong moderation team, a collective sense of community, and usage of moderation tools. Additional

community-characteristics we plan to explore are the accessibility of content mentioned in Section 4.2, how prevalent rules are (i.e., are the rules posted somewhere easy to find), reputation thresholds for contributors (i.e., requiring a certain amount of karma on Reddit to post within a subreddit), requiring user verification to participate (e.g., r/Conservative requires users to prove they are conservative before participating), and nature of subreddit content (i.e., does a community primarily share text, images, or other media). Understanding these causal factors is our primary focus for the future as it has practical implications for governing online communities and their ability to withstand crises. We consider this paper to be the first-step towards developing a conceptual framework for community resilience. Though limited and exploratory, evidence from our analyses demonstrates the promise in this line of research and its implications for broadening our understanding of resilient and successful online communities.

## 6 CONCLUSION

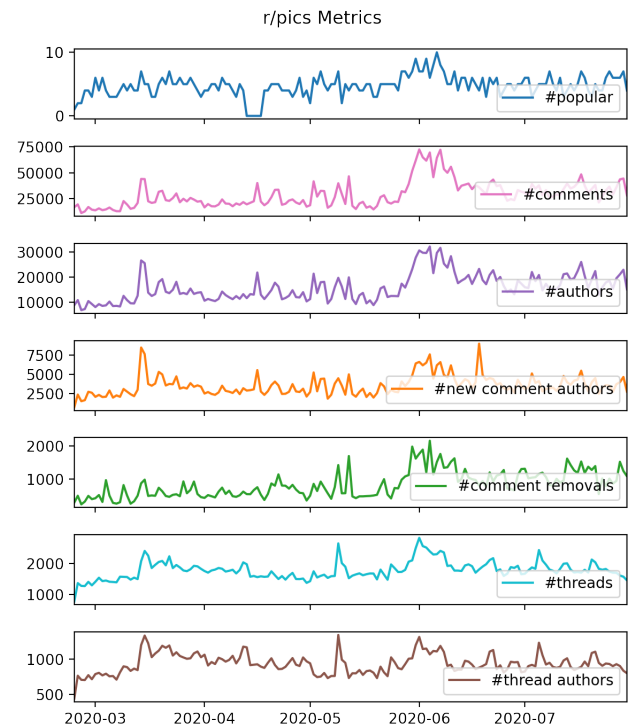
In this paper, we examined thread-level and community-level effects of threads hitting r/popular (the default feed on Reddit) in order to study how different communities handle sudden spikes in attention. We found that r/popular threads were disruptive due to the elevated number of comments, authors, and comment removals. Additionally, we observed trends relating subreddit size to the impacts of individual r/popular threads measured through six community-level metrics. There were also significant variations found between similar-sized subreddits and subreddits containing similar types of content. These variations present an opportunity for future work to explore causal factors influencing a community’s ability to withstand spikes in disruptive activity during crisis events—the concept we referred to as community resilience. Further investigation into these causal factors has practical implications for online governance and fostering resilient and healthy communities. We believe that resilience is another pillar in the success metrics for online communities, similar to how stability and ability to rebound from crises are desirable in offline communities.

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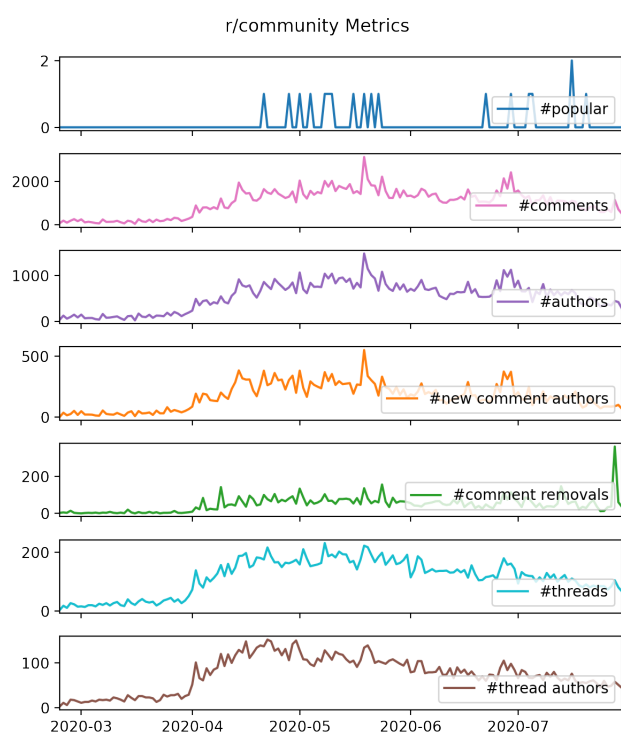
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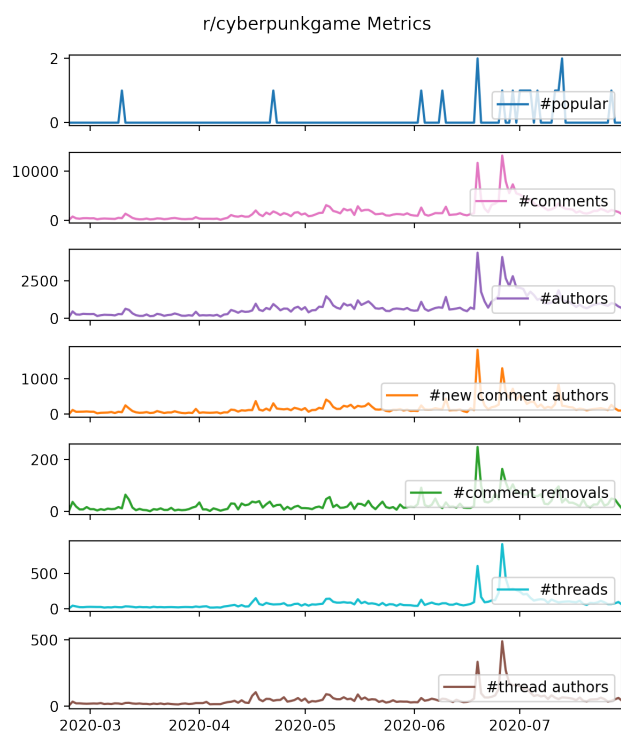
## A APPENDIX



**Figure 1: An example of the metrics generated for each subreddit using the comment and thread data collected. Each graph shows the daily counts for each attribute listed.**



(a) r/community



(b) r/cyberpunkgame

**Figure 3: Comparing subreddit metrics between two similar-sized subreddits: r/community and r/cyberpunkgame.**



**(a) r/Baking thread (ID: fpc085) that reached r/popular.**



**(b) r/Baking thread (ID: fpquw6) that did not reach r/popular.**

**Figure 2: An example of a matched pair produced from propensity score matching used in the Wilcoxon signed-rank test.**